INCH-POUND

MS27749E
27 November 2003
SUPERSEDING
MS27749D
29 Apr 1985

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PDT MAGNETIC LATCH, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 15 NOVEMBER 2002. NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

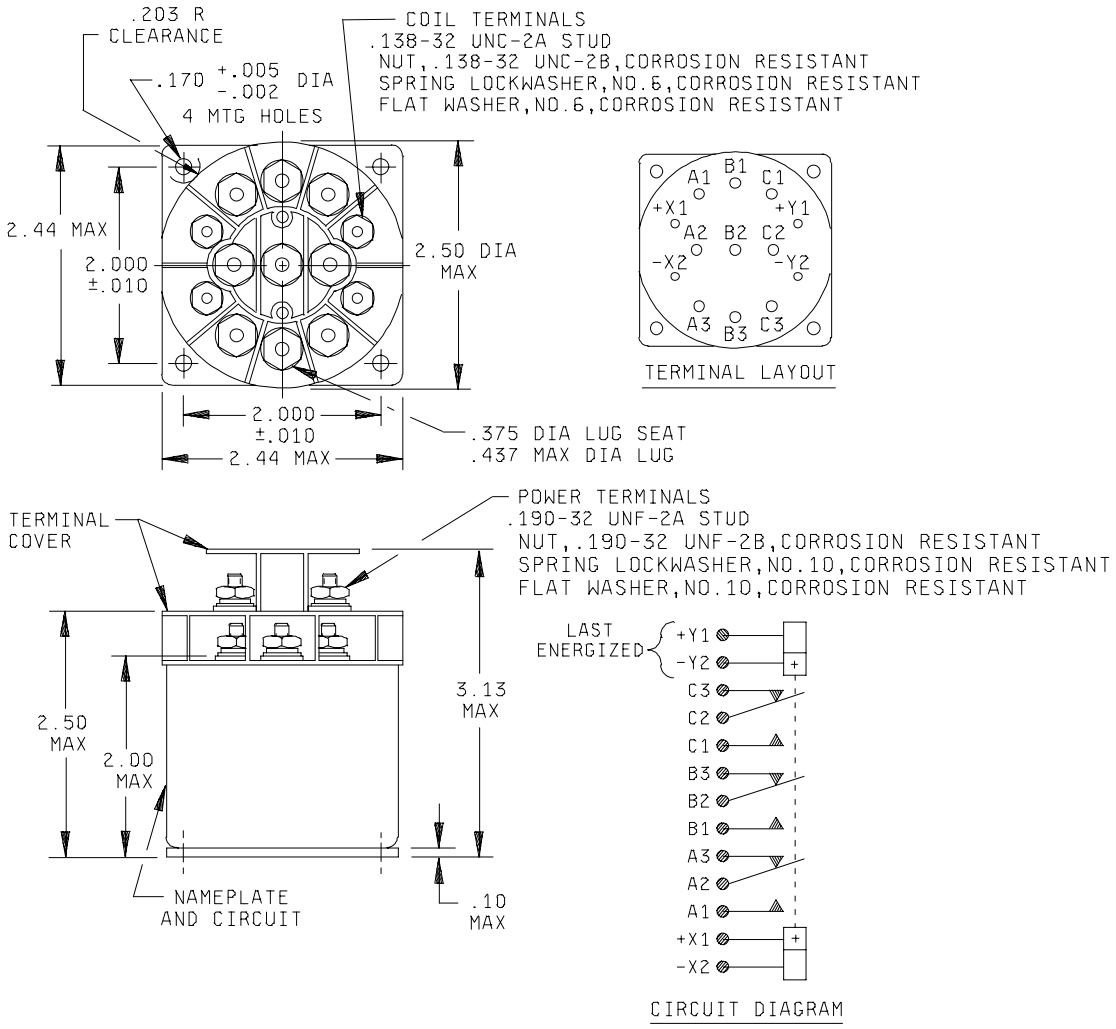



FIGURE 1. Dimensions and configuration.



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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .005 | 0.13 |
| .010 | 0.25 |
| .10 | 2.5 |
| .170 | 4.32 |
| .203 | 5.16 |
| .375 | 9.52 |
| .437 | 11.10 |
| 1.91 | 48.5 |
| 2.000 | 50.80 |
| 2.44 | 62.0 |
| 2.45 | 62.2 |
| 2.5 | 63.5 |
| 3.13 | 79.5 |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
3. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
4. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
5. Relay is magnetically latched in both positions.
6. Coils are not to be energized simultaneously (unless connected series aiding).
7. Metric equivalents are given for general information only.

| MS part number | Type | Coil | Terminal type | Mounting | Max weight in pounds |
|---|---|---|---|---|---|
| MS27749-1 | I | dc | Stud | Plate | .875 |
| MS27749-2 | I | dc | Stud | Plate | .875 |

1/ Weight includes terminal barriers.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Contact data:

Load ratings: See table I.

Initial contact voltage drop: 0.150 volt.

After life test: 0.175 V.

Overload current: 50 amperes dc; 80 amperes ac.

Rupture current: 60 amperes dc; 100 amperes ac.

Coil data: See table II.

Duty rating: Continuous.

Electrical data:

Insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric withstanding voltage:

Sea level:

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 | 1,000 |
| Aux contacts | N/A | N/A |
| All other points | 1,250 | 1,000 |

Altitude:

| | 80,000 ft | 300,000 ft |
|---|---|---|
| Coil to case | 350 | 500 |
| Aux contacts | N/A | |
| All other points | 350 | 500 |

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock g-level: 200 g's.

Duration: 6 ms.

Max duration contact opening: 10 µs.



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Vibration - sinusoidal:

G-level 30 g's.

Frequency range 10 - 3,000 Hz.

Vibration - random:

Applicable specification: MIL-STD-202, method 214.

Test condition: IG.

Duration: 15 minutes each plane.

Acceleration 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: 0.18 pound maximum.

Part or Identifying Number (PIN): MS27742- (dash number from table II).

Qualification by similarity: See MIL-PRF-6106.

TABLE II. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | | 400 Hz | 60 Hz | |
| Resistive 2/ | 50 | 25 | 25 | | | 25 | | | | 25 | | | | |
| Inductive | 10 | 12 | 12 | | | | | | | | | | | |
| Inductive | 20 | | | | | 15 | | | | 15 | | | | |
| Motor | 50 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Lamp | 50 | 5 | 5 | | | 5 | | | | 5 | | | | |
| Transfer load | --- | | | | | --- | | | | --- | | | | 3/ |
| Mechanical life reduced current | 200 | 6 | 6 | | | 6 | | | | 6 | | | | |
| Mixed loads | Applicable per specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ For full rated load, temperature, and altitude, use number 12 wire or larger. Relays shall be mounted so that mounting bracket temperature is limited to +135°C maximum.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.



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Operating characteristics.

| Part no. MS 27749- | Coil data | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max at +25°C | | Max pick-up voltage | | | Drop out volt-age 2/ | Oper -ate 3/ | Rel-eas e 4/ | Contact Bounce | | | |
| | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res ±10% | Volts | Amp | Nor-mal 2/ | High temp test | Cont cur-rent test | | | | NO | NC | NO | NC |
| 1 | X1,X2 | 28 | dc | 200 | 29 | .190 | 18 | 20 | 21 | N/A | 35 | --- | 3 | 3 | --- | --- |
| 2 | Y1, Y2 | 28 | dc | 200 | 29 | .190 | 18 | 20 | 21 | N/A | --- | 35 | 3 | 3 | --- | --- |

1/ Contact transfer time @ rated voltage, 1.8 milliseconds minimum.
2/ Over the temperature range..
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE II. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 50 | 50 | 50 | | | 60 | | | | 60 | | | | |
| Inductive | 20 | 20 | 20 | | | | | | | | | | | |
| Inductive | 50 | | | | | 60 | | | | 60 | | | | |
| Motor | 50 | 20 | 20 | | | 40 | | | | 40 | | | | |
| Lamp | 50 | 10 | 10 | | | 15 | | | | 15 | | | | |
| Transfer load | 10 | | | | | 12.5/ 60 | | | | 12.5/ 60 | | | | 2/ |
| Mechanical life reduced current | 200 | 7 | 7 | | | 14 | | | | 14 | | | | 2/ |
| Mixed loads | 50 | 1 | 1 | | | 5 | | | | 5 | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsychronized ac power supplies at rating indicated. 12.5 amps for -1, 60 amps for -2.

APPLICATION, NOTES:
1. Examination of product, external parts shall be performed in accordance with MIL-PRF-6106, except the case temperature shall be limited to 150°C maximum.
2. Strength of terminals, shall be performed in accordance with MIL-PRF-6106, except it shall be tested at room ambient temperature only.



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ENVIRONMENTAL CHARACTERISTICS

| | |
|---|---|
| Temperature range | 55°C to +71°C. |
| Max altitude rating | 50,000 ft. |
| Shock G-level | 50g. |
| Duration | 6 ms |
| Max duration contact opening | 10 μs. |
| Vibration: | |
| Sinusoidal: | |
| G-level | 10 g |
| Frequency range | 70-2,000 Hz |
| Nonoperate | |
| G-level | 15 g |
| Frequency range | 20-2,000 Hz |
| Acceleration | 15 g |

ELECTRICAL CHARACTERISTICS

| | |
|---|---|
| Insulation resistance, initial | 100 megohms. |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | N/A | N/A |
| All other points | 1,500 V rms | 1,150 V rms |

Dielectric strength (altitude).

| | 50,000 ft |
|---|---|
| Coil to case | 500 V rms |
| Aux contacts | N/A |
| All other points | 700 V rms |

| | |
|---|---|
| Max contact drop initial | .150 volt. |
| After life test | .175 volt. |
| Overload current | 125 A dc; 115/200 A ac; |
| Rupture current | 150 A dc; 115/200 Vac; 400 Hz 500 A |
| Duty rating | Continuous. |



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Conformance inspection: Performance of groups B and C tests may be suspended at the discretion of the qualifying activity.

Qualification by similarity: See MIL-PRF-6106.

NOTES

Referenced documents. In addition to MIL-PRF-6106, this specification sheet references the following documents. (Government documents are available on line at http://assist.daps.dla.mil/quicksearch or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

STANDARDS

Department of Defense

| | | |
|---|---|---|
| MIL-STD-202 | - | Electronic and Electrical Component Parts |
| MIL-STD-461 | - | Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment |

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-1214-01)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil.



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